



Hands-On With the Neo4j Graph Platform Session 4



Class Setup

- Aura -unpause your instance **(Highly recommended)**
- Neo4j Desktop - Start your DB

Class Agenda

- Labeled Property Graph (LPG) Components 🙌
- What is a graph database? 🙌
- What is Neo4j? 🙌

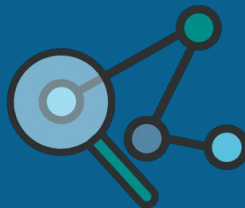
➤ Common Graph Use Cases



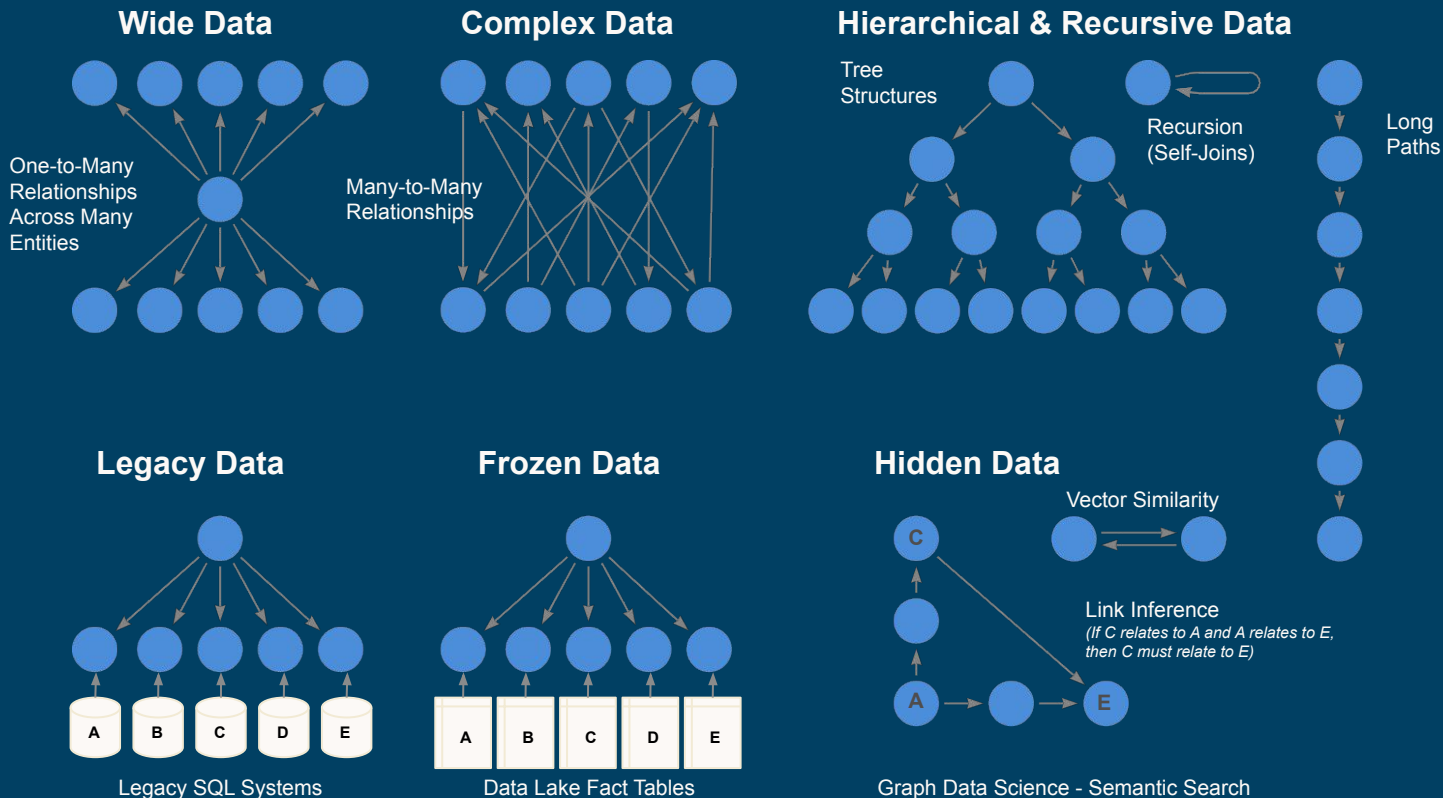
- Graph modeling -Part 1 🙌

- Graph modeling -Part 2 🙌
- Importing CSV into Neo4j
- Constraints and Indexes
- Intro to Graph Data Science
- Graph Data Science workflows

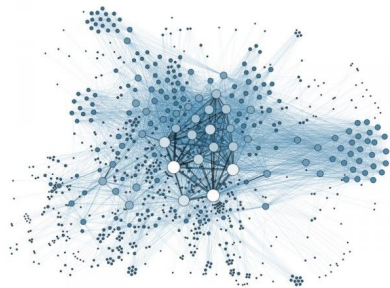
🙌 Hands-On



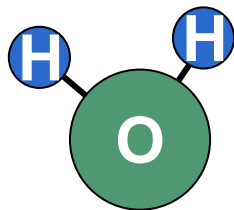
Graphs Naturally Handle Complex Data



Graphs Are Everywhere!



The Internet



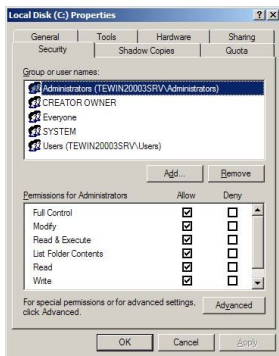
Chemistry



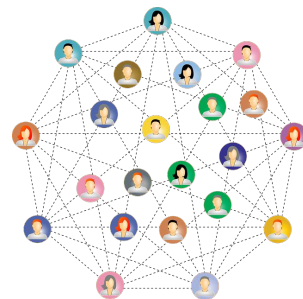
Public
Transit



Supply Chains



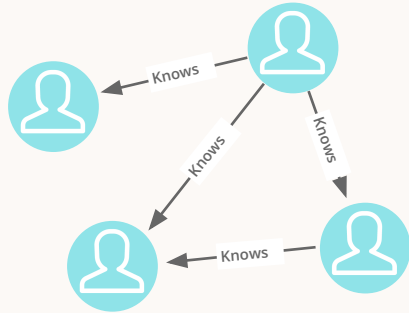
ActiveDirectory
& LDAP



Social Networks

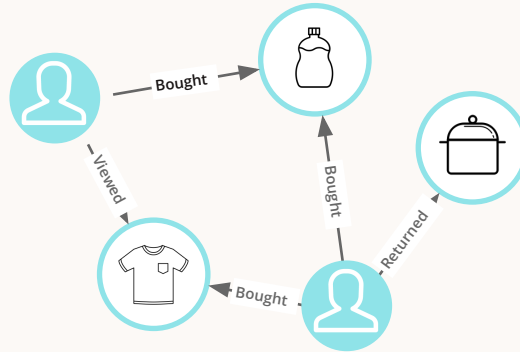
Data Connections establish *Context*

Data Connections Reveal **Paths, Patterns, Dependencies**



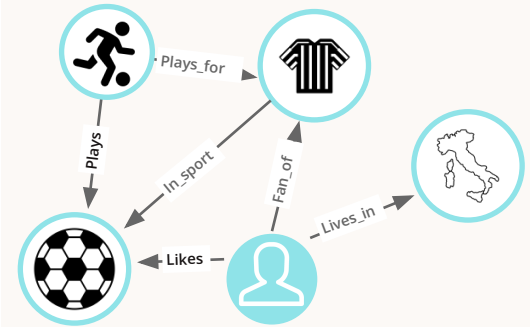
Networks of People

Employees, Customers,
Suppliers, Partners,
Influencers, etc.



Transaction Networks

Risk management,
Supply chain, Orders,
Payments, etc.



Knowledge Networks

Enterprise content,
Domain specific content,
eCommerce content, etc



Common Graph Use Cases

Gartner's 2020 predictions for graph technology:

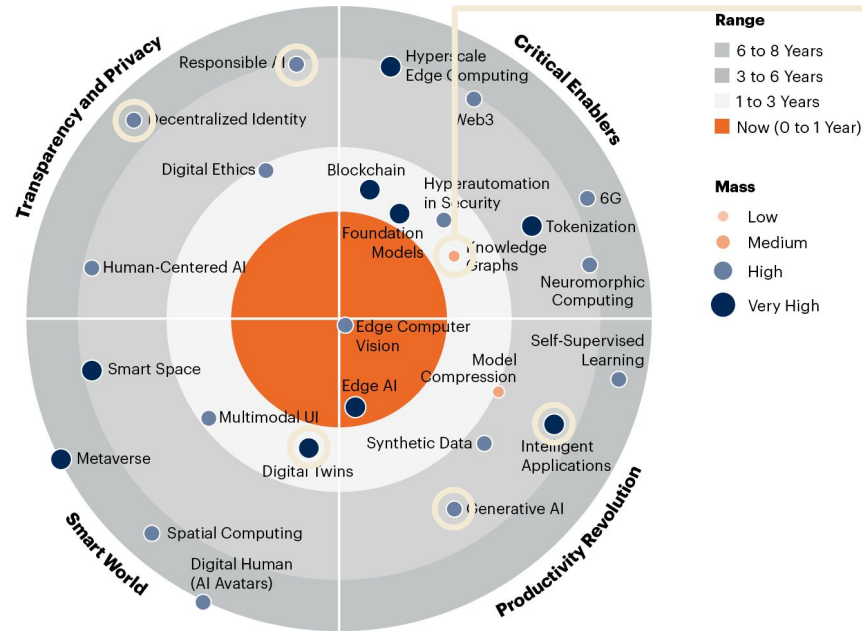
“By 2025, graph technologies will be used in 80% of data and analytics innovations...”

Gartner

Top 10 Trends in Data and Analytics, 11 May 2020, Rita Sallam, et al.

Knowledge Graphs are a Critical Business Enabler

2023 Gartner Emerging Technologies and Trends Impact Radar

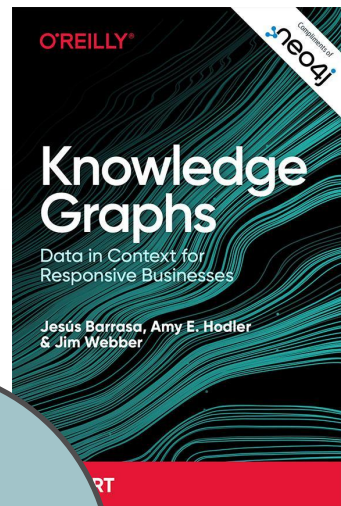
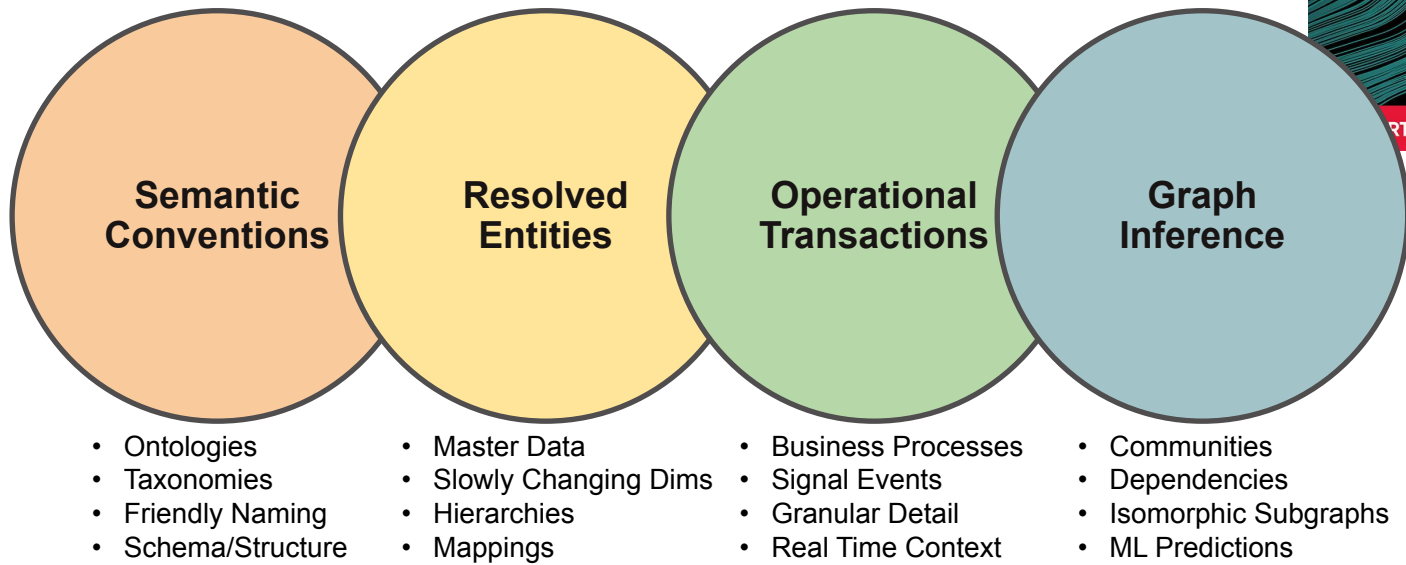


Knowledge Graphs

- Digital Twins
- Intelligent Applications
- Decentralized Identity
- Responsible AI
- Generative AI

What is a Knowledge Graph?

A knowledge graph combines consistent business **semantics**, **entities** extracted and unified from source data, detailed **transactional flows**, and in-graph **analytics/inference** for decision support.



Knowledge Graph Entities

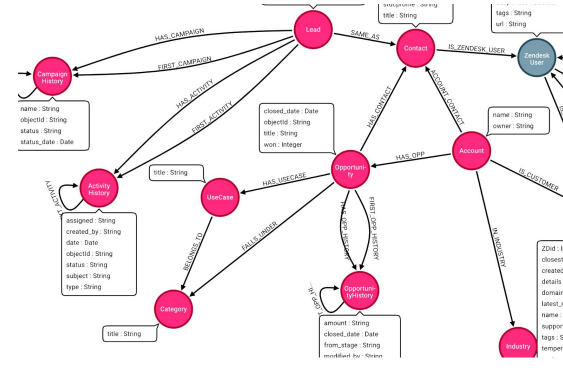


Enable various use cases across enterprises

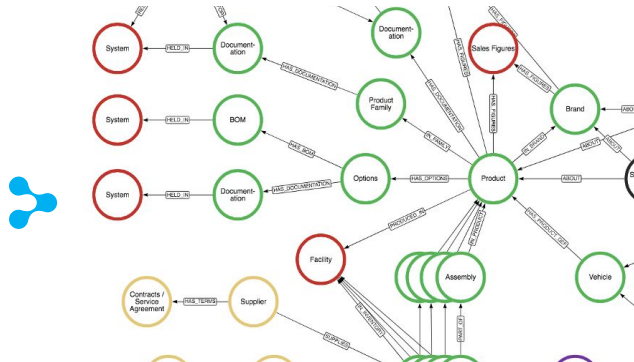
Customer Journey graph



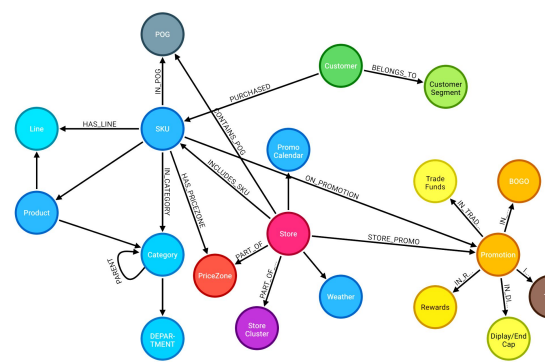
Recommendations graph



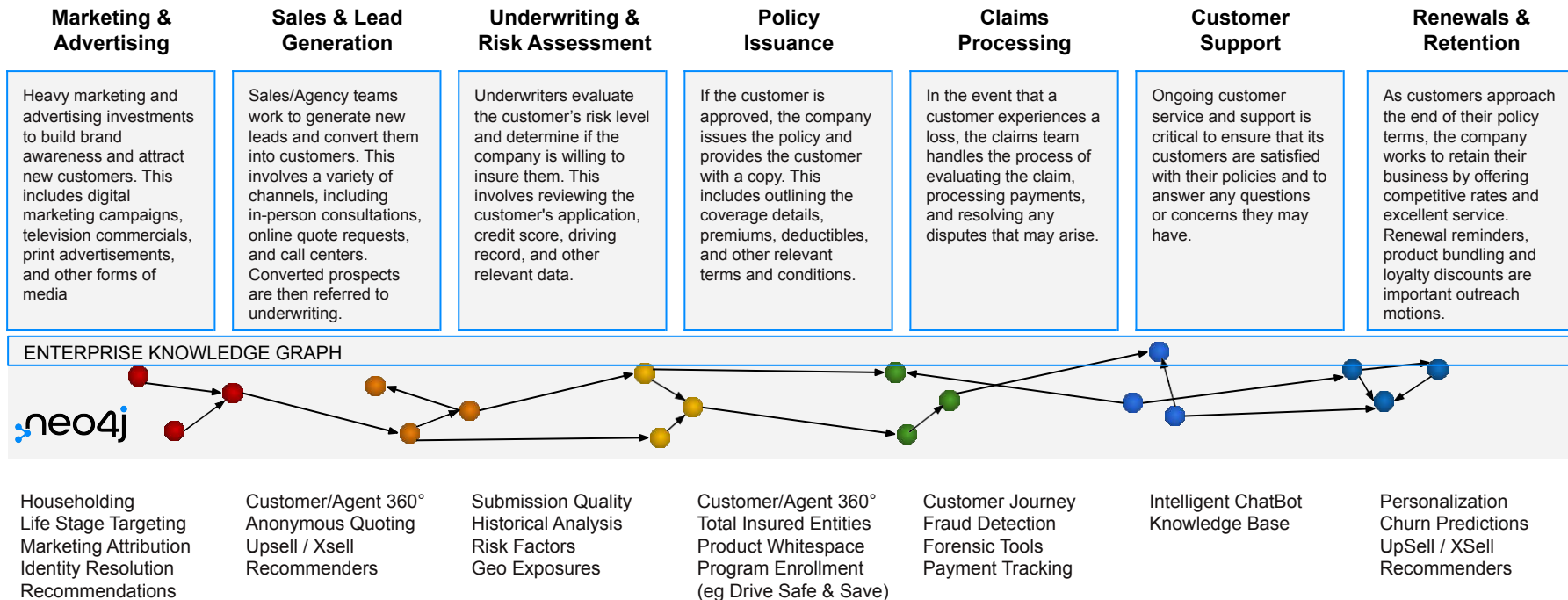
Underwriting graph



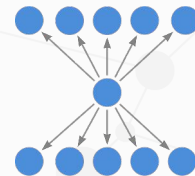
Producer graph



Graph Use Cases by Insurance Processes



Example Graph Schema: Insurance Agent 360°



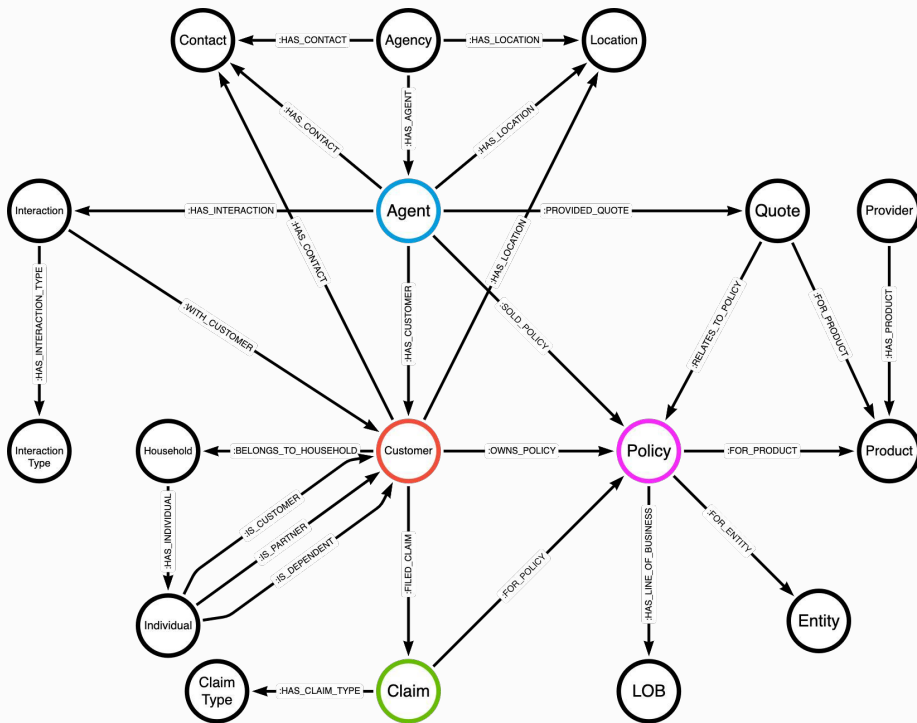
Graph Design

Provides the agent a 360° view of customer activity.

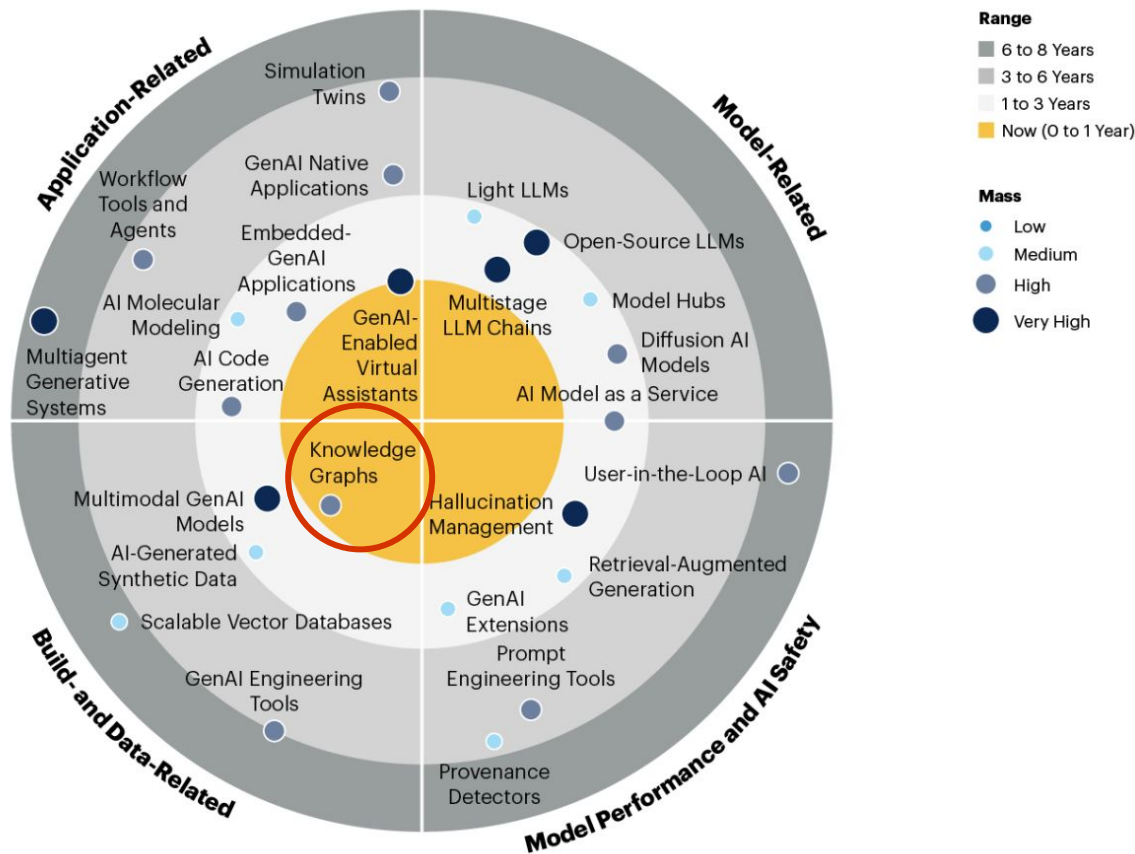
Connects historical data (policies, claims, quotes) with real-time interactions (customer support, web events, mobile).

Handles householding complexity.

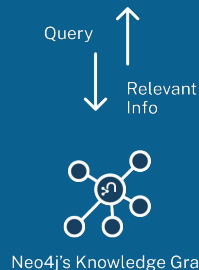
Supports customer analytics and next-best-action product recommendations.



Impact Radar for Generative AI



Gartner identifies Knowledge Graphs as a key enabler for Generative AI



Graph RAG

Neo4j's Knowledge Graph



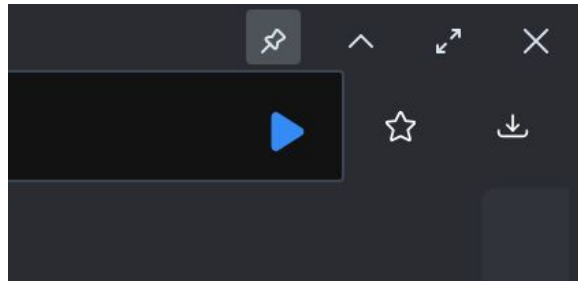
Hands On Time!

Now let's get hands on!

Type `:guide hands-on` into your Neo4j Browser command line

```
neo4j$ :guide hands-on
```

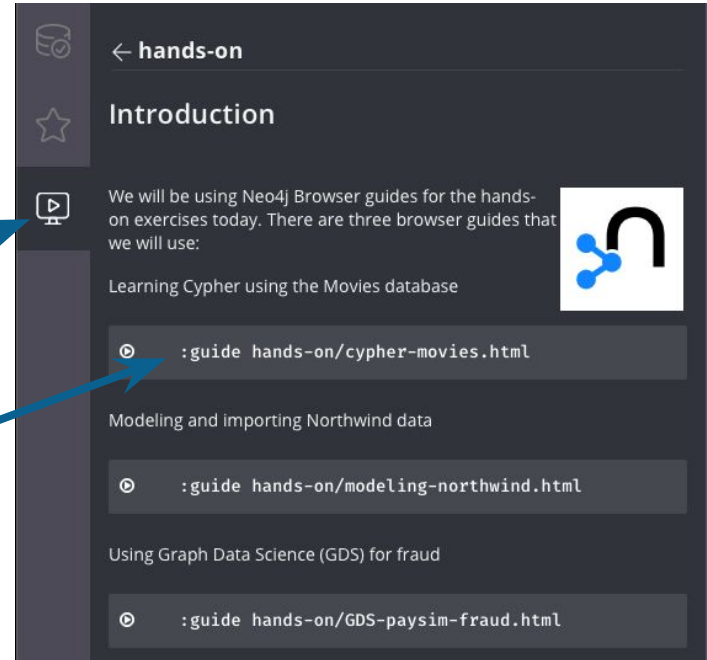
Pin the window -button in the top right corner



1)



2)



Additional Query Exercises

:guide hands-on/cypher-movies.html

- ~~1. Filtering queries using the WHERE clause (13 query exercises)~~
~~hands-on/cypher-exercise_2.html~~
- ~~2. Working with patterns in queries (6 query exercises)~~
~~hands-on/cypher-exercise_3.html~~
- ~~3. Working with Cypher data (COLLECT, COUNT, maps)~~
~~hands-on/cypher-exercise_4.html~~
- 4. Controlling query processing (WITH)**
hands-on/cypher-exercise_5.html
- 5. Controlling returns returned (RETURN, DISTINCT, ORDER BY LIMIT)**
hands-on/cypher-exercise_6.html

Cypher Manual

<https://neo4j.com/docs/cypher-manual/current/introduction/>

Cypher Cheat Sheet

<https://neo4j.com/docs/cypher-cheat-sheet/5/auradb-enterprise/>